

University of Toronto, Department of Electrical and Computer Engineering

ECE 1387 - CAD for Digital Circuit Synthesis and Layout

September 2025

Instructor: Jason Anderson
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Office Hours: After class or by appointment
Course Web: **Use Piazza link on course Quercus page**

Prerequisites: ECE 1388 (VLSI Design Methodology), or CSC 2410 (Algorithms in Graph Theory), or Permission of instructor. Main prerequisites: **programming skills in C/C++**, including data structures, and a **desire to learn** about CAD for digital circuits!

Lecture: Fridays from 5:00-7:00 PM in WB-130, starting September 5

Papers/Readings: Available in PDF on Piazza website.

Evaluation:	Assignments	50% (3 or 4)	Paper	25%
	Exercises	15% (2 or 3)	Quizzes (2)	10%

Assignments: Programming implementations of CAD problems such as placement, routing, and technology mapping using optimization strategies such as simulated annealing, dynamic programming, integer linear programming (ILP), and branch and bound, and illustrated using computer graphics.

Exercises: Hands-on experience with CAD tools such as ABC (UC Berkeley), Mockturtle (EPFL), VPR (Auto Place & Route), and hMetis (Partitioning).

The Paper: A critical assessment of work in a subset of the field (chosen in consultation with the instructor) based on 3 to 4 papers.

Quizzes: Short 10 minute quizzes at the end of selected lectures. The dates will be announced on the Piazza site one week in advance.

TENTATIVE Lecture and Assignment Schedule – Fall 2025

#	Date of Friday Lecture	Lecture Topic	Assignment/ Exercise Handed Out	Assignment/ Exercise Due
1	Sept 5	Introduction, Overview	Paper	
	Sept 12	NO CLASS (Jason is Traveling)		
2	Sept 19	Routing	Assignment 1 – Routing	
3	Sept 26	Timing-Driven Routing		
4	Oct 3	Placement		
5	Oct 10	Placement (Analytical Techniques)	Assignment 2 – Analytical Placement	Assignment 1
6	Oct 17	Placement (Simulated Annealing)	Exercise 1 – VPR placement and routing	
7	Oct 24	Partitioning or SAT Solvers (Branch and Bound)	Assignment 3 – B&B as applied to partitioning or Boolean SAT	Assignment 2
	October 31	NO CLASS Fall Reading Week		
8	Nov 7	Timing Analysis and Slack Allocation		Exercise 1
9	Nov 14	Partitioning (FM/Multi-Level: hMetis)	Exercise 2 – Partitioning Using hMetis or FM	Assignment 3
10	Nov 21	Technology Mapping / Logic Synthesis (Dynamic Programming)	Exercise 3 – Technology Mapping Using Dynamic Prog in ABC (UC Berkeley)	Exercise 2
11	Nov 28	High-Level Synthesis 1 (Intro + Scheduling)	Assignment or Exercise – LegUp High-Level Synthesis or ILP (Integer Linear Programming)	Exercise 3
12	Dec 5	High-Level Synthesis 2		
	Dec 19			Final Assignment/ Exercise

NOTE: You must consult with the instructor on your paper topic by mid-November.